

DISK-TECTIVE WORK: THE LINK BETWEEN INNER DISK GAS AND TERRESTRIAL PLANET FORMATION

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Abstract

Rovibrational carbon monoxide (CO) emission lines provide information about the dynamics and properties of the gas environment in the inner regions of protoplanetary disks, where terrestrial planets are forming. We present a subset of ~ 200 Keck-NIRSPEC spectra of young stars with disks, previously analyzed for their column densities, temperatures, and emitting area. We compare our spectra with spatially-resolved ALMA continuum emission images of the disks to search for a link between the larger outer disk substructure and the inner disk gas properties. We classify disks into six substructures: smooth, cavity, rings, spirals, arc, and streamer. We have found that disks with multiple rings have a CO detection rate of only 54.5% as compared to 71.6% of the ALMA imaged disks. We find no correlation between stellar properties and substructure nor any difference between the spectral or physical properties of the CO emission lines detected in rings versus other substructures. This implies that the decreased CO detection rate is solely correlated with the rings substructure.

Introduction

While it is suspected that the properties of inner protoplanetary disks have an effect on the planets that form in them, the extent to which disk features and chemistry affect the planets remains unclear. By studying disks, we have learned about the formation of planets born from these disks (e.g. Ormel & Klahr (2010)) and how the disk chemistry is linked to their development (Jiang et al. 2023). In continuing such study, we will potentially discover what initial conditions are needed to form a habitable planet.

Rovibrational carbon monoxide (CO) traces the hot gas environment of the inner regions of disks by emitting light at infrared wavelengths. CO can provide information about the dynamics and properties of the inner disk atmosphere. Examining rovibrational emission lines allows us to probe up to the outer edge of the inner disk where it is emitting at temperatures of just a few hundred K. Analyzing the inner disk (< 5 AU) is important for understanding both disks and planet formation, as it is the section of disk that interacts with the star, controlling accretion and ejection of stellar material. This same area is where terrestrial planets are expected to form (Hayashi et al. 1985).

Substructure of the dust within protoplanetary disks has been closely studied by the Disk Substructures at High Angular Resolution (DSHARP) team via the close examination of the mm-dust of 20 disks (Andrews et al. 2018). The team found the sample to contain mostly ringed disks, several spiral arms, and small asymmetrical arcs, with no obvious correlation to the stellar host properties (Andrews et al. 2018). The DSHARP teams work paved the way for other high resolution disk observations, many of which have been used in this study to classify disk morphologies (e.g. Yamaguchi et al. (2024), Cieza et al. (2021)).

A few correlations between inner disk chemistry and system characteristics have already been discovered. Salyk et al. (2011) found a correlation between H₂O linefluxes and the mid-infrared continuum flux, as well as other parameters such as stellar luminosity, mass, and accretion rate. Banzatti et al. (2022) detected the first CO Keplerian rings at < 2 AU and the first H₂O rovibrational lines at $5\ \mu\text{m}$. Banzatti et al. (2022) was also able to relate their CO spectroscopy to disk inclination. Our work is focused on further probing this link between the inner and outer disks by relating inner disk CO properties to large-scale dust substructures in the outer disk.

Methods

Our sample consists of 186 stars with known protoplanetary disks. It contains both low-mass (T Tauri) and high-mass (Herbig Ae/Be) stars that cover a wide range of spectral types, as well as disks across SED classification type (Class I, II, III, and transition disks).

Our spectra were obtained with Keck-NIRSPEC (McLean et al. 1998), which is a high-resolution infrared spectrometer on the Keck II telescope. The observations derive from observing runs spanning from 2001-2019 and form a large Keck-NIRSPEC survey of protoplanetary disks. Spectra were analyzed for their CO (1–0), CO (2–1), and ¹³CO emission lines, from this we gathered linefluxes (the strength of emissions at expected CO wavelengths) continuums (the expected background radiation of each star found from blackbody curves), and line widths (the impact of the doppler effect on our spectra). Slab modelling was used to derive column densities, temperatures, and emitting areas for each disk with sufficient CO emission lines.

We used ALMA images of disks as opposed to SPHERE images because the continuum emissions were more commonly imaged for our sample than the scattered light. All the substructure was observed in millimeter-sized dust grains. A total of 69 disks within our sample were imaged this way (Andrews et al. 2018; Yamaguchi et al. 2024; Cieza et al. 2021). After obtaining a spatially resolved image of a target disk from the literature, we sorted each disk into one of six categories: smooth, cavity, rings, spirals, arc, and streamer. The criteria we used for the first five morphologies are explained in Andrews (2020). Streamer was the only new substructure, which we defined as having an inflow from an arm extended beyond the disk into the envelope, indicative of a younger disk. We disregarded edge-on disks ($\text{incl} > 80^\circ$) as well as shadowed disks.

Results

CO vs Stellar Properties

We first tested correlations between the stellar properties of ringed disks and their CO. There was no correlation between stellar type and CO. Both the T Tauri stars and Herbig Ae/Be stars in our sample had very similar CO detection rates, of 74% and 70% respectively although our sample consisted of only 10 Herbig and 54 T Tauri stars. The substructures were evenly spread across each of the spectral types showing no correlation between the two. The majority of our sample consists of K and M type stars. We had the least B and F type stars, but both still showed a range of substructures

CO vs Substructure

Our first exploration of the link between substructure and rovibrational CO emissions was detection rates among various substructures. Within our sample of 69 continuum emission imaged disks, we find that the majority of the disks are either smooth (no substructure) or contain rings. The disks with rings exhibited a much lower rate of CO detection when compared to the entirety of the imaged sample, while smooth disks and disks with cavities were within 10% of the rest of the sample as shown in Fig 1). All the other substructures had less than 10 imaged objects, so they were not highlighted as a subject of this study.

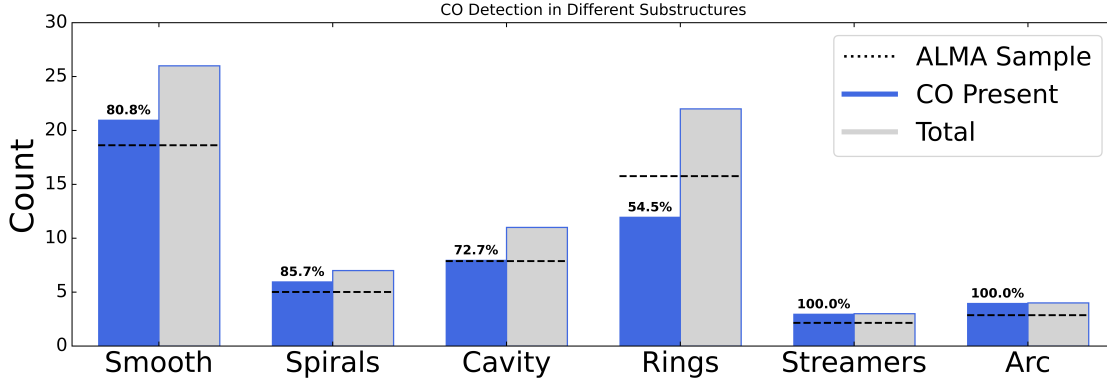


Fig. 1.— The CO Detection rate of our sample, separated by substructure. The dashed line shows the percent value of the CO detection rate of the ALMA sample.

CO vs Retrieved Properties

There was no difference in CO (2–1) detection or ^{13}CO detection as compared to our standard CO (1–0) detection rate as seen in Fig 2.

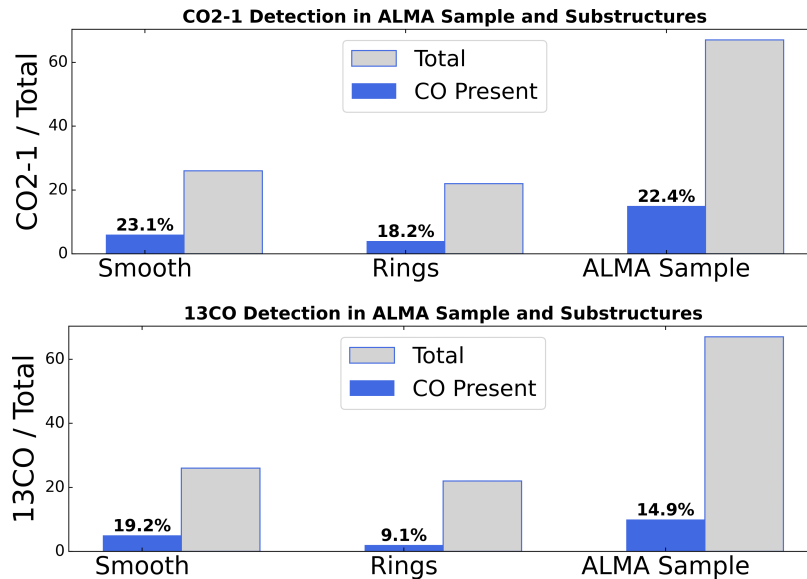


Fig. 2.— The CO detection rates of CO (2–1) and ^{13}CO , respectively of our imaged sample.

There were some trends between the retrieved properties of the CO and the substructures. Smooth disks were found to have a lower median emitting area (Fig. 3), this is likely due to $\sim 90\%$ of the smooth disks in our sample being compact disks ($R_{\text{dust}} < 50$ AU) as compared to the majority of our ringed sample being extended disks. Smooth disks had a greater median column densities, explained by the absence of gaps in the disk. The temperature medians and spreads were also very similar as shown in Fig. 3. We also found that disks with a higher total star-disk system mass, as compiled by Manara et al. (2023), were more likely to have substructure in their disks, Fig 4.

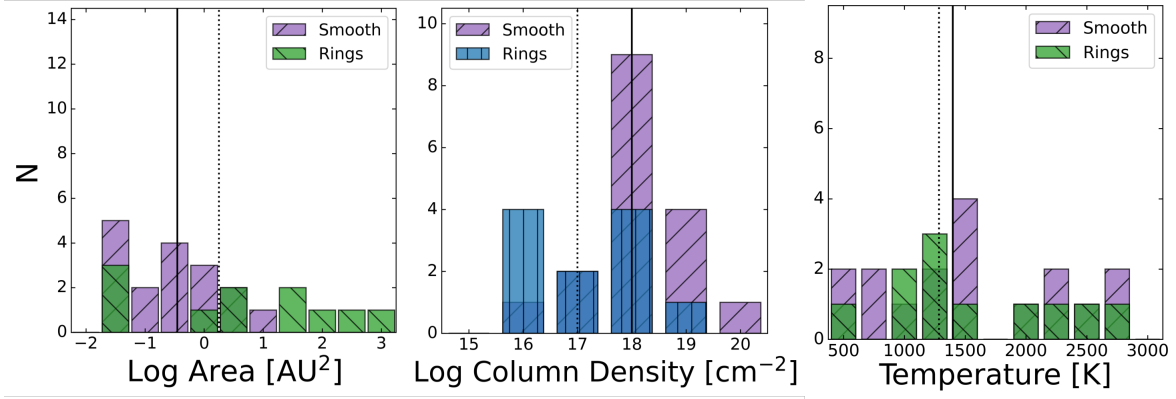


Fig. 3.— The retrieved properties of our sample, from left to right: emitting area, column density, and temperature. The medians of the smooth disks are represented by a solid line, and ringed disk medians are represented by a dotted line.

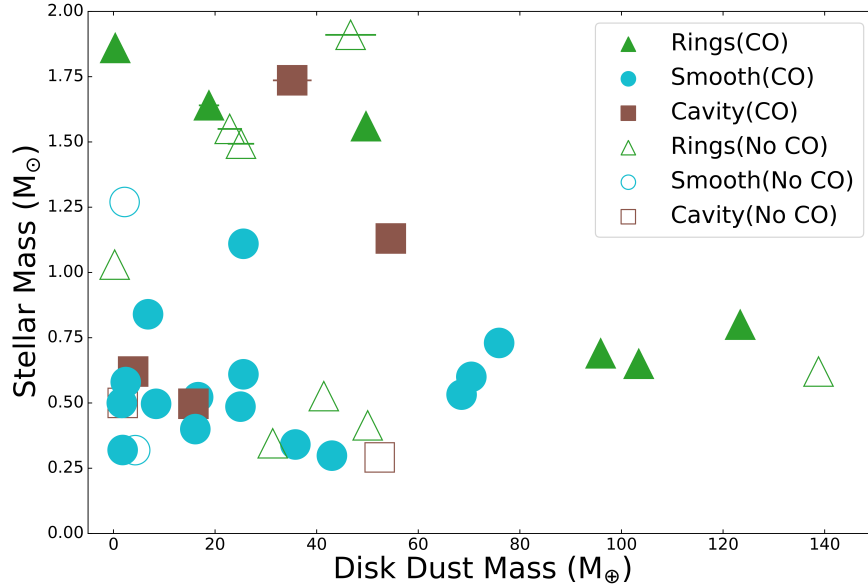


Fig. 4.— Stellar Mass ($M_{\odot}$) against Disk Dust Mass ($M_{\oplus}$) with substructure and CO markers.

Summary and Conclusions

We built a data repository of substructure categorizations for ALMA images. We compiled detailed information from the literature of each of our 186 target protoplanetary disks including data about the host star, and disk parameters from literature. We analyzed previous Keck-NIRSPEC spectra and calculated line fluxes, continuums, and linewidths for CO (1–0), CO (2–1), and ^{13}CO emission lines for all of our targets. We compared our analysis of protoplanetary disk substructures to their spectroscopy and discovered that ringed disks have a lower rate of CO (1–0) detection in their inner disk compared to disks without substructure. We also uncovered potential trends relating substructure to other disk characteristics, such as system mass, column density, and dust radius, though further study is required.

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